THREE EQUIVALENT DESCRIPTIONS OF I-PROJECTIVITY

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Projective and free modules, and the fact that a map factoring through a projective module factors through a free one. the author cites Lemma lemma-ideal-factor-through-projective at proof lines 17772-17774. Gives the equivalence of the first two conditions.
- The class of a short exact sequence in Ext^1 and its vanishing on projective modules. used at proof lines 17774-17780. Provides the direction from factoring to annihilation.
- Annihilation of an extension class by an ideal element as a factorisation statement. used at proof lines 17780-17783. Converts the annihilation condition back into the factoring condition for the third implication.

Source reference. Original lemma 0G92: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 20287–20287. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 0G90: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 20285–20285. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let R be a ring. Let $I \subset R$ be an ideal. Let M be an R-module. The following conditions are equivalent*

(1) *for every $a \in I$ the map $a : M \to M$ factors through a projective R-module,*
(2) *for every $a \in I$ the map $a : M \to M$ factors through a free R-module, and*

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(3) $\mathrm{Ext}^1_R(M, N)$ *is annihilated by* I *for every* R*-module* N.

Proof. The equivalence of (1) and (2) follows from Lemma 0G90. If (1) holds, then (3) holds because $\mathrm{Ext}^1_R(P, N)$ for any N and any projective module P. Conversely, assume (3) holds. Choose a short exact sequence $0 \to N \to P \to M \to 0$ with P projective (or even free). By assumption the corresponding element of $\mathrm{Ext}^1_R(M, N)$ is annihilated by I. Hence for every $a \in I$ the map $a : M \to M$ can be factored through the surjection $P \to M$ and we conclude (1) holds. □